Smaller models. Stronger evidence.

A Minitron-inspired study under an 8 GiB GPU budget

Qwen2.5-3B → physically smaller students

18 recovery runs · 3 seeds · CPU and CUDA evidence

RESEARCH QUESTION

Does the least-damaging cut recover best?

Compare immediate damage with equal-budget recovered quality.

Test whether the answer changes with compression severity and metric.

Judge models by quality, parameters, memory and measured execution cost.

PREREGISTERED DESIGN

A controlled experiment—not a model-size demo

One installed, pinned 3.086B teacher; no paid API or new model download.

Two sizes: 32 layers / FFN 9,600 and 28 layers / FFN 8,192.

Random versus informed pruning × 3 seeds; six matched CE controls.

Each run: 196,608 input tokens and 192 optimizer updates.

Physical pruning with an exact removal recipe

Depth: remove internal contiguous blocks; preserve boundary layers.

Informed depth: calibration NLL; informed FFN: coupled L2 magnitude.

FFN: slice gate/up rows and matching down columns—not masks.

Every layer/channel ID is saved; save/reload shape tests passed.

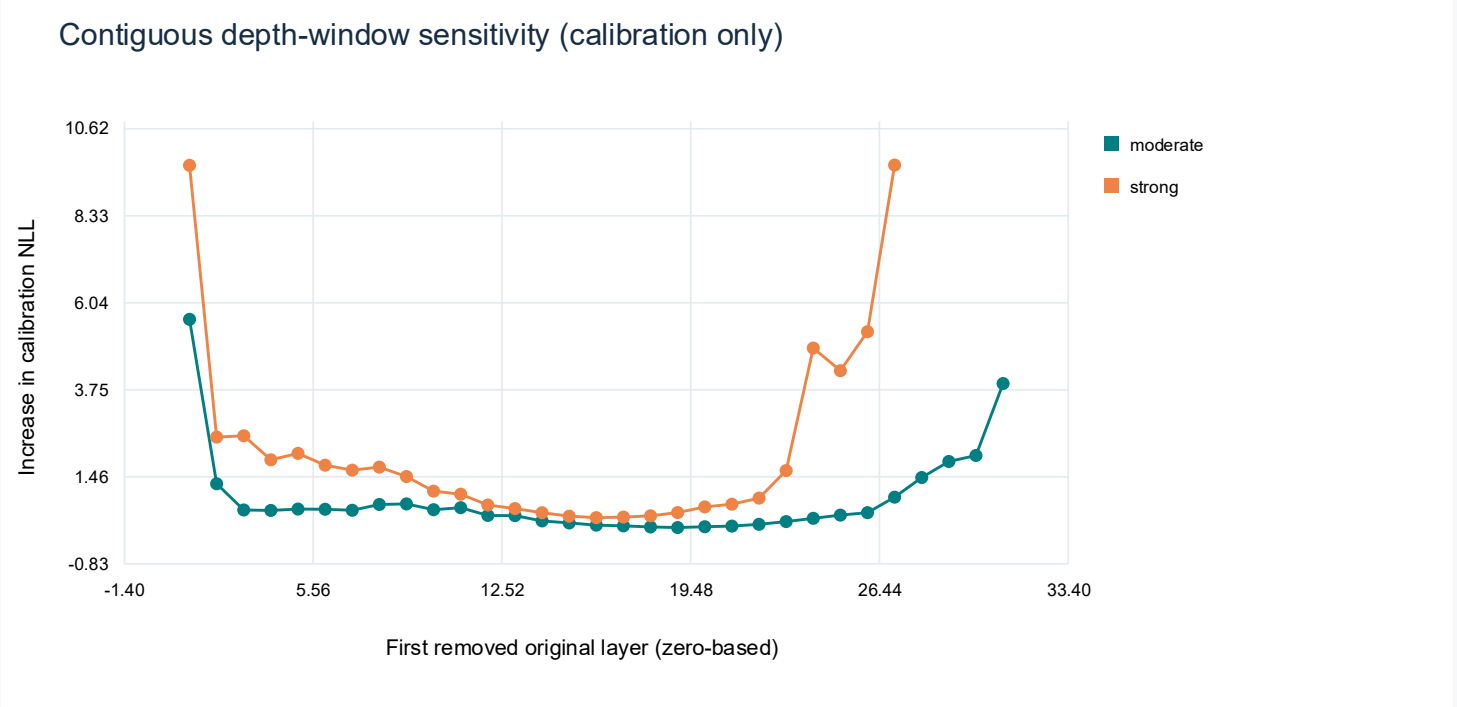
CALIBRATION ONLY

Sensitivity is measured before recovery

32 fixed calibration blocks; test data does not rank structures.

Different depth windows have different immediate damage.

FFN score-distribution map and exact channel selections are included.



Measured local data. Raw timings and per-seed results are retained.

Full-vocabulary KD without two large GPU models

Teacher target shard → unload teacher → recover students.

Forward KL at T=2 with T^2 scaling; KL-only versus CE-only.

Rank-8 q/v LoRA, FP16 frozen base, FP32 adapters, checkpointing.

Atomic checkpoints each update; bounded disk and thermal guards.

MEASURED THREE-SEED RESULTS

Compression–quality trade-offs, not a universal winner

moderate/random/kl: PPL 32.0 ± 2.4; PIQA 70.5%

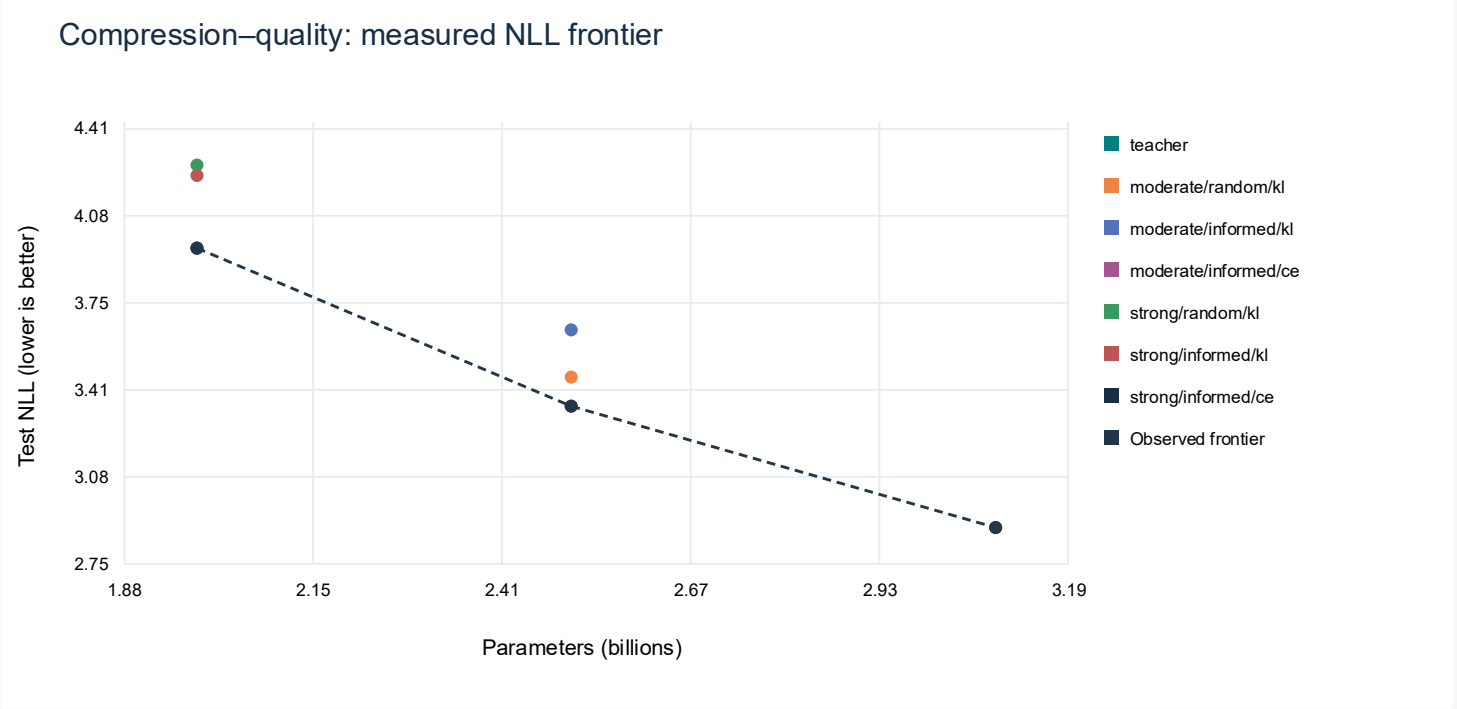
moderate/informed/kl: PPL 38.3 ± 0.2; PIQA 72.3%

moderate/informed/ce: PPL 28.6 ± 0.0; PIQA 72.0%

strong/random/kl: PPL 72.7 ± 13.3; PIQA 60.7%

strong/informed/kl: PPL 69.1 ± 0.8; PIQA 65.8%

strong/informed/ce: PPL 52.3 ± 0.2; PIQA 65.3%



Measured local data. Raw timings and per-seed results are retained.

CONTROLLED SCIENTIFIC COMPARISON

Recovery and the isolated distillation ablation

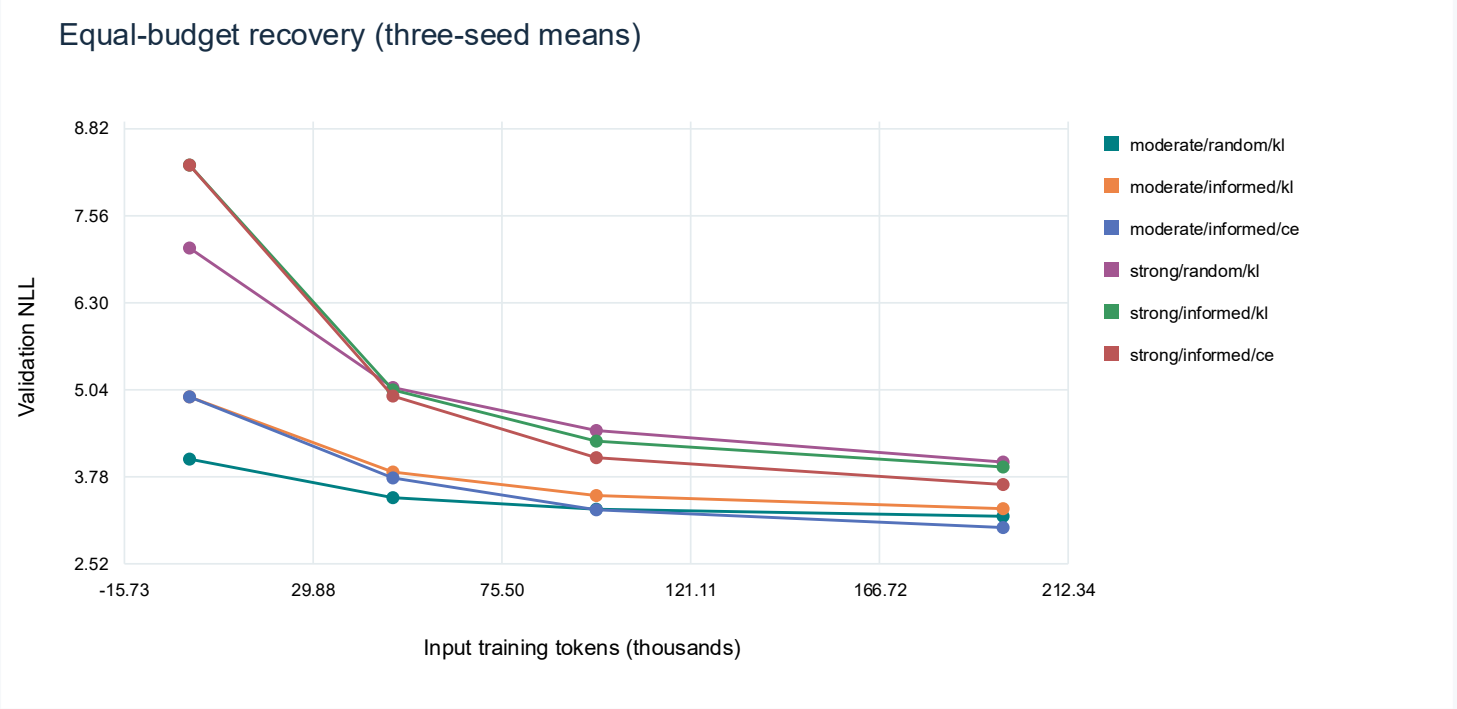
moderate: informed minus random KL → ΔNLL +0.181, ΔPIQA +1.8pp

moderate: KL minus CE informed → ΔNLL +0.292, ΔPIQA +0.3pp

strong: informed minus random KL → ΔNLL -0.040, ΔPIQA +5.2pp

strong: KL minus CE informed → ΔNLL +0.278, ΔPIQA +0.5pp

Matched CE recovery had lower test perplexity at both compression levels.



Measured local data. Raw timings and per-seed results are retained.

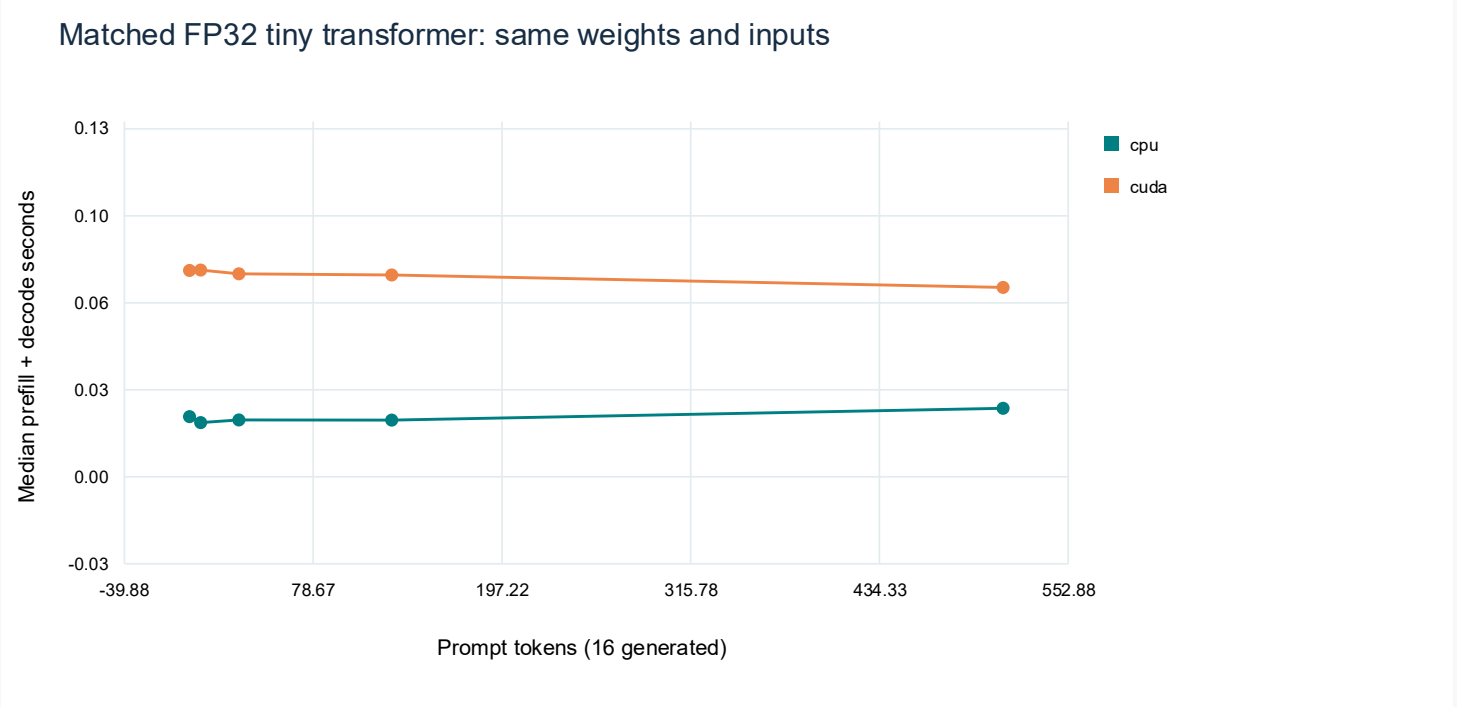
CPU and CUDA both execute—not just a device flag

Complete tiny-transformer train → prune → distill on both devices.

Three-seed FP32 reference; same-weight device benchmark.

Real teacher and two students: matched quality inputs and PIQA scoring.

CPU FP32 versus CUDA FP16 is labelled device + precision.



Measured local data. Raw timings and per-seed results are retained.

MEASURED WORKLOAD SCALING

Where is GPU deployment practically beneficial?

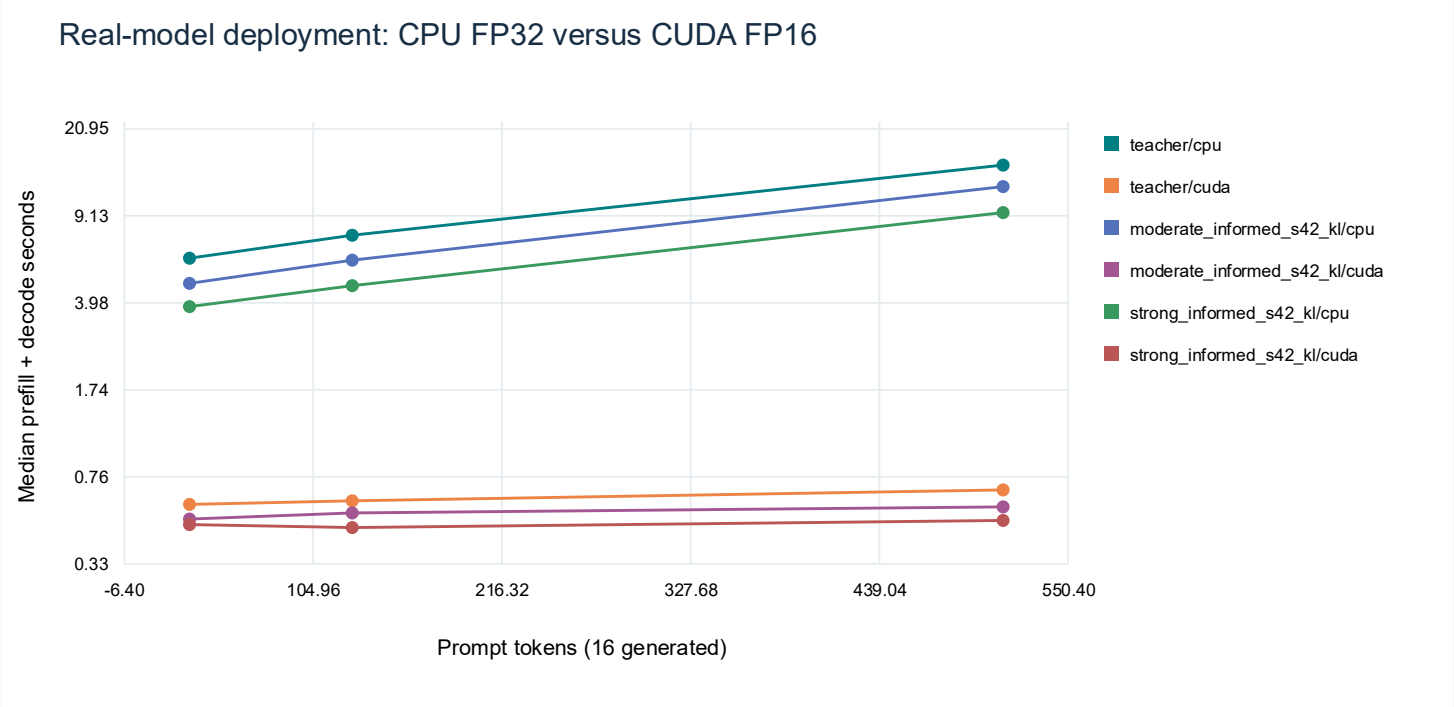
teacher: 32→16: 10.4×, 128→16: 12.6×, 512→16: 22.1×

moderate: 32→16: 9.4×, 128→16: 11.1×, 512→16: 21.2×

strong: 32→16: 8.0×, 128→16: 10.0×, 512→16: 18.8×

2 warmups · 5 timings · medians + IQR · explicit placement

Loading, preprocessing, transfer and model execution are separate.



Measured local data. Raw timings and per-seed results are retained.

ENGINEERING EVIDENCE

A recoverable run with a visible resource budget

Max main training allocation: 5.19 GiB.

Max sampled training GPU temperature: 65°C.

Peak target shard payload: 2.30 GiB.

Hash-verified checkpoints, corruption fallback, STOP / RESUME.

All final adapters retained; one merged representative exported.

What this project establishes—and what it does not

Immediate NLL ranking persisted in 4/6 seed pairs—not universally.

Executed CPU/GPU protocol, Pareto analysis, sensitivity and objective controls.

Not an official NVIDIA-scale reproduction or production chatbot.

Limits: one family/machine, 3 seeds, short context, custom PIQA, no teacher correction.

Code, raw evidence, recipes, adapters, report and reproduction guide.